

University of Luxembourg, Luxembourg City, Luxembourg

July 1, 2026

Dear Hiring Committee,

Having recently graduated as an AI & Data Science Engineer with a Master's degree in relevant fields, I am thrilled to apply for the Doctoral Researcher position in Computer Vision at the University of Luxembourg. The opportunity to work on vision-language-action models in autonomous driving, in collaboration with POST Luxembourg, aligns perfectly with my research interests and goals.

With a strong background in programming skills and proficiency in languages such as Python and PyTorch, I am confident that my expertise will be an asset to your team. My recent experience working on the "Cardiovascular Medical Digital Twin" project has equipped me with a unique set of skills that can be applied to this PhD position.

One achievement that stands out is my work on the MedicalMambaTS, a linear-time state space model for sub-millisecond bedside inference. This project demonstrated my ability to design and implement efficient algorithms for high-stakes applications, which aligns with the requirements for developing vision-language-action models in autonomous driving. The low-latency capabilities of MambaTS can be applied to real-time computer vision tasks, ensuring timely decision-making in autonomous systems.

Another relevant achievement is the PIDeepONet, a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd. This project showcased my ability to integrate physical constraints into neural networks, which is crucial for developing robust and reliable models in computer vision.

I am excited about the prospect of joining the University of Luxembourg as a Doctoral Researcher and contributing to the development of innovative solutions in Computer Vision. Please find attached my CV, academic references, transcripts, and the one-page summaries of both my Engineering and Master's theses. Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui